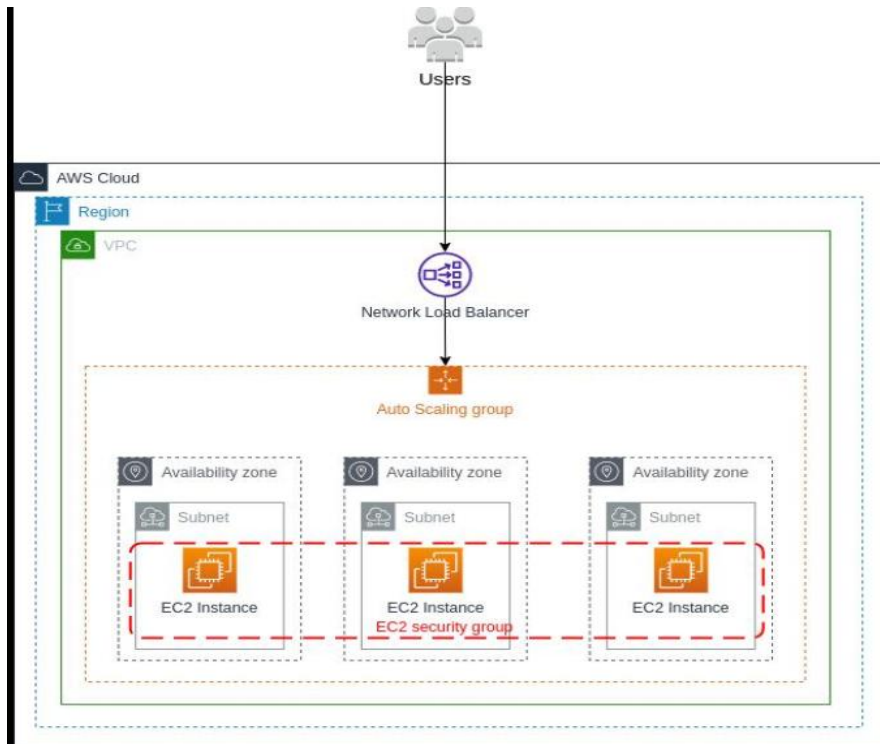


Name : Vivek Chaudhari

Project : 2

FluxScale: High-performance scalable web app infrastructure



Project Overview (Simple Explanation)

FluxScale is a scalable web application architecture built using AWS that can handle heavy traffic with low latency. It automatically adjusts resources based on demand and ensures high availability using load balancing and scaling techniques.

⚙️ AWS Services Used & Why

1. 🟡 Amazon EC2

Why used:

EC2 instances act as servers where your web application runs.

Script / Explanation:

- Launch multiple EC2 instances
- Deploy your web app (HTML, Java, Python, etc.)
- These instances handle user requests

👉 Think: “Actual machines running your app”

2. Network Load Balancer

Why used:

To distribute incoming traffic efficiently with **ultra-low latency**.

Script / Explanation:

- Accepts user requests
- Distributes traffic across EC2 instances
- Works at Layer 4 (TCP/UDP) → faster than ALB

👉 Think: “Traffic controller for high-speed requests”

3. Amazon EC2 Auto Scaling

Why used:

Automatically increases or decreases EC2 instances based on traffic.

Script / Explanation:

- Monitors CPU usage / traffic
- Adds instances when load increases
- Removes instances when load decreases

👉 Think: “Auto expansion and shrink system”

How This Project Works (Flow)

1. User sends request
 2. Network Load Balancer receives request
 3. NLB forwards request to EC2 instances
 4. Auto Scaling ensures enough instances are running
 5. User gets fast and reliable response
-

Summary

FluxScale is a highly scalable and low-latency web application architecture using EC2, Network Load Balancer, and Auto Scaling. It ensures efficient traffic handling, automatic scaling based on demand, and high availability, making it suitable for high-performance applications like gaming, streaming, or real-time systems.

NLB Server Running

EC2 > Instances

EC2

Dashboard

AWS Global View

Events

▼ Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Capacity Manager

▼ Images

AMIs

AMI Catalog

▼ Elastic Block Store

Volumes

Snapshots

Instances (1/8) Info

Last updated 2 minutes ago

Connect

Instance state

Actions

Launch instances

Saved filter sets

Choose filter set

Find Instance by attribute or tag (case-sensitive)

< 1 >

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4
☐		i-08ff5ee4c8b318d68	Terminated	t3.micro	–	View alarms +	ap-south-1c	–
☐		i-0b9931085609c6835	Running	t3.micro	3/3 checks passed	View alarms +	ap-south-1c	ec2-15-206
☐		i-0b1cb44d4b68d37b6	Running	t3.micro	3/3 checks passed	View alarms +	ap-south-1c	ec2-13-201
☐	web-server	i-0b6a7440a9c9aa8c1	Running	t3.micro	3/3 checks passed	View alarms +	ap-south-1a	ec2-3-110-1
☐		i-0808aa4ec6ad5d8d9	Terminated	t3.micro	–	View alarms +	ap-south-1a	–
☐		i-0a315e33bfdab6d43	Running	t3.micro	3/3 checks passed	View alarms +	ap-south-1a	ec2-3-108-6
☐		i-0312c79e1f4c5ea75	Running	t3.micro	3/3 checks passed	View alarms +	ap-south-1a	ec2-15-206
☒	nlb-server	i-0e71e03168fe7e5a9	Running	t3.micro	3/3 checks passed	View alarms +	ap-south-1a	ec2-13-232

i-0e71e03168fe7e5a9 (nlb-server)

Details | Status and alarms | Monitoring | Security | Networking | Storage | Tags

▼ Instance summary Info